1 Connected Substitutes

Consider a demand model with utility given by the random utility function

$$v_{ijt} = v_j(x_t, \xi_t, p_t; \theta_{it})$$

which is often parametrized via the linear random coefficients model as $v_{ijt} = x_{jt}\beta_{it} - \alpha_{it}p_{jt} + \xi_{jt} + \epsilon_{ijt}$, though we do not require this restriction for the connected substitutes definition. As usual, x are the observed product characteristics, p the price, ξ the unobserved product characteristics, and ϵ the idiosyncratic preferences.

Let $\chi_t = (x_t, p_t, \xi_t)$ denote the characteristics of the products and/or markets. The market shares (choice probabilities) are given by

$$s_{jt} = \sigma_j(\chi_t) = \mathbb{P}(\underset{k \in \mathcal{J}}{\operatorname{argmax}}\, v_{ikt} = j | \chi_t), \qquad j = 0,...,J.$$

Also partition $x_{jt} = (x_{jt}^{(1)}, x_{jt}^{(2)})$ and define the linear indices

$$\delta_{jt} = x_{jt}^{(1)}\beta_j + \xi_{jt}, \qquad j = 1,...,J.$$

This indices are intended to affect the distribution of utilities. In Berry and Haile (2014), we condition on $x_{jt}^{(2)}$ for the identification argument.

We are now ready to define connected substitutes.

Definition 1.1 (Connected Substitutes)**.** *Let λ_t denote either $-p_t$ or δ_t. Goods (0,...,J) are **connected substitutes** in λ_t if both of the following hold.*

1. *$\sigma_k(\delta_t, p_t)$ is nonincreasing in λ_{jt} for all $j > 0, k \neq j$.*
2. *At each $\sigma_k(\delta_t, p_t)$, for any nonempty $\mathcal{K} \subseteq 1,...,J$, there exists $k \in \mathcal{K}$ and $l \notin \mathcal{K}$ such that $\sigma_l(\delta_t, p_t)$ is strictly decreasing in λ_{kt}.*

Point 1 requires that goods be weak gross substitutes in δ_t, p_t. For example, as price increases, the market share will decrease. Point 2 requires some strict substitution, enough to justify treating $\mathcal{J}$ as the relevant choice set. That is, as the market share for one product increases, another decreases due to substitution effect.